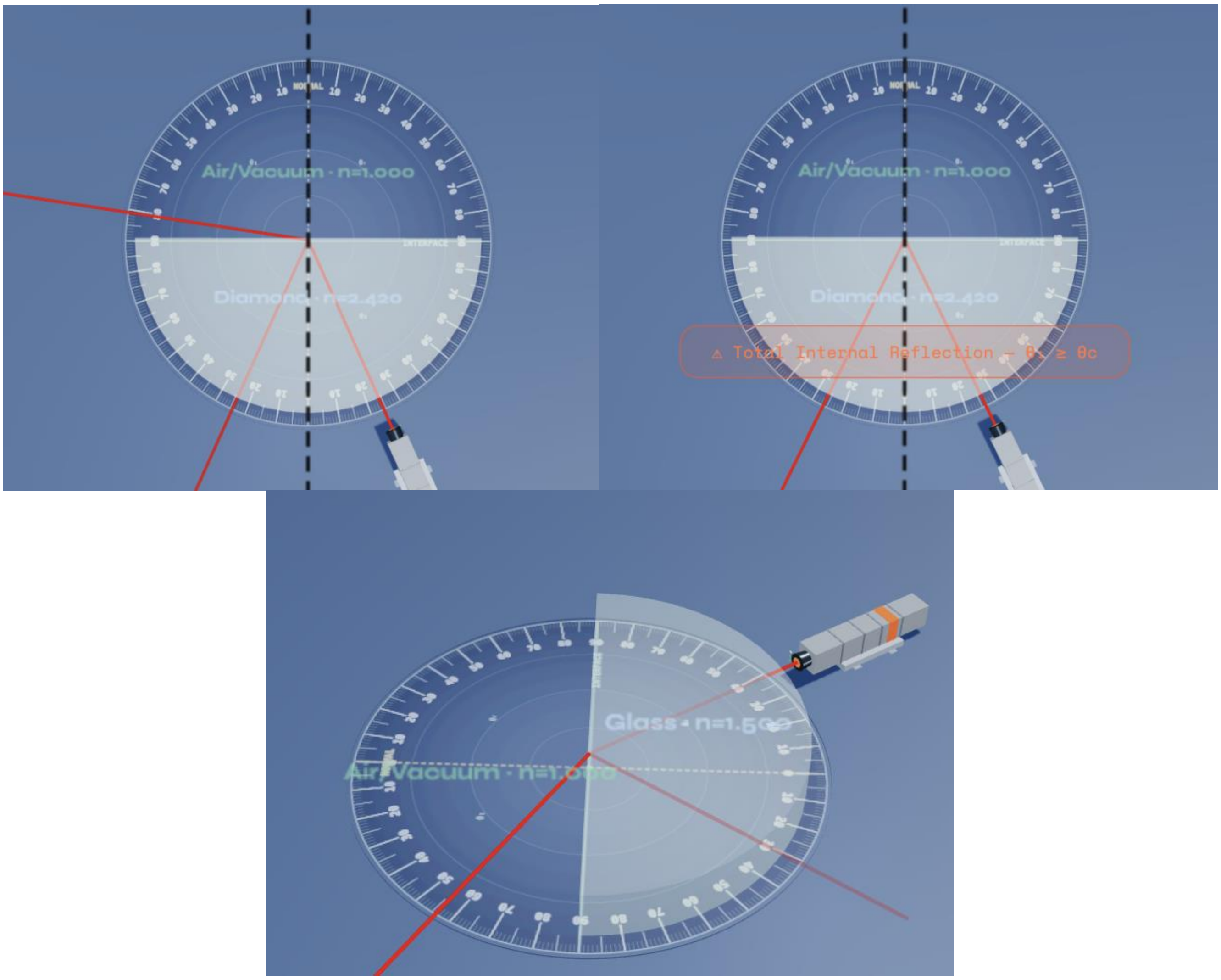
LAB 2: TOTAL INTERNAL REFLECTION AND THE CRITICAL ANGLE

1 OBJECTIVES

- To investigate the behavior of light when transitioning from an optically dense medium (n_1) to a less optically dense medium (n_2).
- To understand the conditions necessary for Total Internal Reflection (TIR) to occur.
- To experimentally determine the critical angle (θ_c) of an unknown mystery semi-circular prism.
- To calculate the refractive index (n_1) of the mystery material based on the measured critical angle.

2 THEORETICAL BACKGROUND

When light travels from an optically dense medium with a higher refractive index (n_1) into a less optically dense medium with a lower refractive index (n_2), the refracted ray bends *away* from the normal line. Consequently, the angle of refraction (θ_2) is always larger than the angle of incidence (θ_1).



As the angle of incidence increases, the angle of refraction reaches $90°$ at a specific incident angle. This threshold angle is defined as the Critical Angle (θ_c):

$$n_1 \sin\theta_c = n_2 \sin 90^\circ$$

Since $\sin 90° = 1$, we can solve for the critical angle:

$$sin\theta_c = \frac{n_2}{n_1}$$

s

If the second medium is air or vacuum ($n_2 = 1.000$), the equation simplifies to:

$$n_1 = \frac{1}{sin\theta_c}$$

For any incident angle greater than or equal to the critical angle ($\theta_1 \geq \theta_c$), no light is transmitted through the boundary. Instead, 100% of the light is reflected back into the dense medium. This phenomenon is called Total Internal Reflection (TIR).

3 EQUIPMENT & SIMULATION SETUP

- Interactive Refraction Lab Simulator (refraction_lab.html)
- Incident Medium (*Medium* 1): Mystery Material semicircular prism (n_1 =???)
- Refracted Medium (*Medium* 2): Air / Vacuum (n_2 = 1.000)
- Light Source: Laser diode set to green wavelength (λ = 532 nm)
- Measurement Aids: Circular protractor scale, TIR Alert banner in the simulator, zoom loupe.

4 EXPERIMENTAL PROCEDURE

Part A: Entering TIR Challenge Mode

1. Open the virtual simulator refraction_lab.html .
2. Locate the Measurements HUD on the right-hand side.
3. Click on the θ_c critical row inside the HUD (indicated by ↗click). This opens the critical angle educational tooltip.
4. Click the blue 🔍 Find the Mystery n button at the bottom of the tooltip.
5. This activates Challenge Mode. In this configuration, the semicircular prism becomes the mystery material (placed on the incident n_1 side) and the refracted medium is set to Air / Vacuum (n_2 = 1.000).

Part B: Pinpointing the Critical Angle

1. Set the camera view to ⊙ Top in the view panel for the most accurate vertical perspective.

2. Slowly drag the laser angle slider (θ_1) upwards, starting from 30.0°.

Note how it bends closer and closer to the horizontal interface (90° from the normal) as you increase the angle.

4. Continue increasing θ_1 in very small increments (use the slider carefully or tap the fine slider adjustment).
5. Stop at the exact angle where the refracted ray completely merges with the flat interface boundary and then vanishes. This threshold is marked by:
   - The reflection coefficient R reaching 1.00 (100% reflection) on the HUD bar.
   - The appearance of the red warning banner: "⚠ Total Internal Reflection — $\theta_1 \geq \theta c$".
6. Record this threshold angle as θ_c. Back off the angle slightly and approach it again to verify your precision to the nearest 0.5° or 0.1°.

5 SCREENSHOT SUBMISSION PLACEHOLDER

[PLACE KEY SCREENSHOT HERE]

Capture your simulation workspace at the exact moment of Total Internal Reflection (TIR).

Your screenshot must clearly display the red TIR warning banner, and the laser aimed precisely at the critical angle, with the refracted beam vanishing on the interface.

6 DATA ACQUISITION & CALCULATION

Table 2.1: Experimental Critical Angle Data

- *Constant:* Refractive Index of Refracted Medium (n_2) = 1.000 (Air)

Trial	Approached Angle from Left (deg)	Approached Angle from Right (deg)	Measured Critical Angle θc (Average)	Calculated Semicircular Prism $n_1 = \frac{1}{sin\theta_c}$
1				
2				
3				
				Mean $n_{1,\,exp}$ = _____

7 MATERIAL ANALYSIS & IDENTIFICATION

Utilizing the same reference list from Lab 1, match your calculated incident index $n_{1,\,exp}$ with its true optical identity.

Table 2.2: Standard Reference Material Index List

Material	Refractive Index (n)	Material	Refractive Index (n)
Fluorite	1.43	Flint Glass	1.65
Fused Quartz	1.46	Sapphire	1.77
Crown Glass	1.52	Cubic Zirconia	2.15
Polycarbonate	1.59	Gallium Phosphide	3.05

- Identified Mystery Material: ________________________
- Reference Refractive Index ($n_{1,\,ref}$): _______________
- Percentage Error Calculation:

$$\%\ Error = \left|\frac{n_{1,exp} - n_{1,ref}}{n_{1,ref}}\right| \times 100\% = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\%$$

8 EVALUATION QUESTIONS

1. Critical Condition Requirement: Why does Total Internal Reflection *not* occur when light travels from air into a glass block ($n_{air} < n_{glass}$)? Explain this using the mathematical formulation of Snell's Law.

2. Real-world Application: Fiber optic communication cables rely completely on Total Internal Reflection. Based on what you observed during this lab, explain why fiber optic cores are manufactured with extremely high refractive indexes, wrapped inside a cladding layer of lower